iSolarCloud Documentation_Lite

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Authorization

Account Login Authorization Process Introduction

If you already have an iSolar Cloud account and can view these power stations on iSolar Cloud, you can directly use this account to obtain a token through the login interface for identity authentication to access other interfaces.

Obtain a Token through The Login Interface

Before calling the interface, you need to call `/openapi/login` interface to get the token:

Return example:

```
POST /openapi/login

Content-Type: application/json
x-access-key: your application's Secret Key
sys_code: 901

{
   "appkey": "your appkey",
   "user_account": "your account",
   "user_password":"your account password"
}
```

Return example:

```
{
  "req_serial_num": "202411114fa4436f9bc807bc966920ec",
  "result_code": "1",
  "result_msg": "success",
  "result_data": {
    "user_master_org_id": "89899046",
    "mobile_tel": "153****5",
    "user_name": "test user1",
    "language": "Chinese",
    "token": "999125867_91bbb35be9f74ccebb0406307ca80874",
    "err_times": "0",
    "user_id": "1",
    "login_state": "1",
```

```
    "disable_time": null,
    "country_name": "China",
    "user_account": "test user1",
    "user_master_org_name": "org-ty1867erni",
    "email": "",
    "country_id": "1"
  }
}
```

Call the API Interface through Token

Then you can take `/openapi/login` the token returned by the interface calls the interface of the open platform:

```
POST/openapi/getPowerStationList

Content-Type: application/json
x-access-key: your application's Secret Key
sys_code: 901

{
    "appkey":"your appkey",
    "token":"token",
    "curPage":1,
    "size":10
}
```

Return example:

```
{
  "req_serial_num":"202411124b514f528da409106b0bec7b",
  "result_code":"1",
  "result_msg":"success",
  "result_data":{
    "pageList":[
      {
        "ps_name":"MN20220720A",
        "online_status":0,
        "latitude":31.86406773478788,
        "description":null,
        "valid_flag":1,
        "grid_connection_status":1,
        "ps_fault_status":3,
        "ps_location":"H2O UNITY Coffee",
        "update_time":"2024-09-02 17:20:37",
        "ps_current_time_zone":"GMT-2",
        "ps_id":342459181,
        "grid_connection_time":20230808152800,
        "connect_type":1,
        "install_date":"2023-08-08 15:27:58",
        "build_status":2,
        "ps_type":4,
        "longitude":117.27770144878463
```

```
    }
  ],
  "rowCount":1
 }
}
```

Login

Post /openapi/login

Login

1. Request Parameters

Request Parameters	Parameter Type	Required	Parameter Example	Parameter Description
user_account	String	Yes	sungrowUser	User account
user_password	String	Yes	pw111111	User Password

```
{
  "user_password":"1qaz@WSX",
  "user_account":"polana7333@tsderp.com"
}
```

2. Return Parameters

Return Parameters	Data Type	Example Value	Description
req_serial_num	String	20240920d9184e7fa2e9a031e5e96f5b	Request Sequence Number
result_code	String	1	Error Code
result_msg	String	success	Prompt Message
result_data	Map	--	Returned Data
result_data.disable_time	String	2024-11-27 13:44:45	Account Lock Date
result_data.email	String	yang@163.com	User Email Address
result_data.err_times	String	0	Wrong Password Entries
result_data.language	String	Chinese	User Language
result_data.login_state	String	1	Login status: -1: The account does not exist 0: the password is incorrect 1: the login is successful 2: The account is locked due to incorrect password 5: The account is locked by the administrator
result_data.mobile_tel	String	13155366666	User Mobile Number
result_data.token	String	503093_fce7a5839e6c46f998952d55c0384774	Returned Token After Successful Authentication
result_data.user_account	String	Autotest	User Account
result_data.user_id	String	503093	User ID
result_data.user_master_org_id	String	527815	User Organization ID
result_data.user_master_org_name	String	Brenergie BV	User Organization Name

result_data.user_name	String	org-bjrdeen0qi	Username
result_data.country_name	String	China	Country Name
result_data.country_id	String	1	Country ID

2.1 Success Examples

```
{
  "req_serial_num":"202411133cea4e6c8ef73fd79f8f82a7",
  "result_code":"1",
  "result_msg":"success",
  "result_data":{
  "user_master_org_id":"530435",
  "mobile_tel":null,
  "user_name":null,
  "language":"English",
  "token":"505120_2a62bd4e737e4e8fabc9caaa97f032c6",
  "err_times":"0",
  "user_id":"505120",
  "login_state":"1",
  "disable_time":null,
  "country_name":"Germany",
  "user_account":"bwq770s46t",
  "user_master_org_name":"M-wise",
  "email":"polana7333@tsderp.com",
  "country_id":"50"
}
}
```

2.2 Failure Example

```
{
  "req_serial_num": "202411133cea4e6c8ef73fd79f8f82a7",
  "result_code": "1",
  "result_msg": "success",
  "result_data": {
  "login_state": "-1",
  "msg": "账号不存在"
}
}
##
```

2.3 Error Code

Error Code	Description
-1	User account does not exist
0	Password Error
1	Login Success
2	Account locked due to incorrect password
5	Account locked by admin

Monitoring

Plant List Information Query

Post /openapi/getPowerStationList

Query a user's plant list information or query the plant list information based on the user's organization ID. If you cannot find the required information, get the ps_id of the plant through this endpoint and call the "real-time plant data endpoint" while passing in the required data measurement point number.

1. Request Parameters

Request Parameters	Parameter Type	Required	Parameter Example	Parameter Description
ps_name	String	No	PlantA	Plant Name (Fuzzy Search)
ps_type	String	No	1,3,4,5,6,7,8	Plant Type (separate multiple values with commas, all values are queried by default): 1: Utility Plant; 3: Distributed PV Plant; 4: Residential PV Plant; 5: Residential Energy Storage Plant; 6: Village Plant; 7: Distributed Energy Storage Plant; 8: Poverty Alleviation Plant; 9: Wind Power Plant; 10: Utility Energy Storage Plant; 12: C&I Energy Storage Plant
share_type	String	No	0,1,2	Share type of the power plant: 1: query the power plant that has the browse permission 2: query the power plant that has the management permission 0: query the power plant that has the own power plant. Use commas (,) to separate multiple types. All types are queried by default
valid_flag	String	No	1,3	The power plant status to be queried is as follows: (1: normal, 2: disabled, 3: connecting) If the node is not reported, the power plant in status 1 is queried. If multiple statuses need to be queried, separate them using commas (,).
org_id	String	No	527815	If querying by organization ID, then pass in the user_master_org_id output parameter returned by the corresponding login endpoint.
curPage	Integer	Yes	1	Page
size	Integer	Yes	10	Page Size
user_id	String	No	506574	User id (The user must be a junior to the login account)
is_self_built	Integer	No	0	Self-built power plant 0- All 1- Self-built Default 0 All

```
{
   "curPage": 1,
   "size": 10
}
```

2. Return Parameters

Return Parameters	Data Type	Example Value	Description
req_serial_num	String	2022122373544aab92527faf91b7addc	Request Sequence Number
result_code	String	1	Error Code
result_msg	String	success	Prompt Message
result_data	Map	--	Returned Data

result_data.pageList	List	--	Paginated Data
result_data.pageList.alarm_count	Integer	0	Alarms
result_data.pageList.build_status	Integer	2	Plant Construction Status (0: Not Started; 1: Under Construction; 2: Connected; 3: Planned; 4: Not Connected)
result_data.pageList.co2_reduce	Map	--	CO_2 reduction Today
result_data.pageList.co2_reduce.unit	String	千克	Unit
result_data.pageList.co2_reduce.value	String	0	Numerical Value
result_data.pageList.co2_reduce_total	Map	--	Total CO_2 reduction
result_data.pageList.co2_reduce_total.unit	String	Ton	Unit
result_data.pageList.co2_reduce_total.value	String	661405	Numerical Value
result_data.pageList.co2_reduce_update_time	String	2024-11-14T16:00:50+08:00	The format of the last update time for co2 reduction today is yyyy-MM-dd'T 'hh :mm:ssXXX
result_data.pageList.connect_type	Integer	1	Grid Connection Type: (1: Full Grid; 2: Self-Generated and Self-Used (Surplus Power); 3: Self-Generated and Self-Used (No Feed Network); 4: Off-Grid)
result_data.pageList.curr_power	Map	--	Real-time Power
result_data.pageList.curr_power.unit	String	TWp	Unit
result_data.pageList.curr_power.value	String	5	Numerical Value
result_data.pageList.co2_reduce_total_update_time	String	2024-11-14T16:00:50+08:00	Last updated time for cumulative co2 emission reduction Format: yyyy-MM-dd'T 'hh :mm:ssXXX
result_data.pageList.curr_power_update_time	String	2024-11-14T16:00:50+08:00	Last update time of the current power format: yyyy-MM-dd'T 'hh :mm:ssXXX
result_data.pageList.description	String	Rooftop power station	About the Plant
result_data.pageList.equivalent_hour	Map	--	Today's equivalent hours
result_data.pageList.equivalent_hour.unit	String	h	Unit
result_data.pageList.equivalent_hour.value	String	8	Numerical Value
result_data.pageList.equivalent_hour_update_time	String	2024-11-14T16:00:50+08:00	Last updated time Format: yyyy-MM-dd'T 'hh :mm:ssXXX
result_data.pageList.fault_count	Integer	0	Fault Count
result_data.pageList.install_date	String	2022-11-07 15:37:40	Plant Creation Time (format: yyyy-MM-dd HH:mm:ss)
result_data.pageList.latitude	Num	41.69395123697499	Longitude
result_data.pageList.longitude	Num	123.18212151404492	Latitude
result_data.pageList.ps_current_time_zone	String	GMT+8	Current Plant Time Zone

result_data.pageList.ps_fault_status	Integer	3	Plant Fault Status (1: Fault; 2: Alarm; 3: Normal)
result_data.pageList.ps_id	Integer	729056	Plant ID
result_data.pageList.ps_location	String	Longtu Rd, Shushan District, Hefei, Anhui, China	Plant Location
result_data.pageList.ps_name	String	PlantA	Project Name
result_data.pageList.ps_status	Integer	0	power plant status 1: online, 0: offline
result_data.pageList.ps_type	Integer	4	Plant Type (separate multiple values with commas, all values are queried by default): 1: Utility Plant; 3: Distributed PV Plant; 4: Residential PV Plant; 5: Residential Energy Storage Plant; 6: Village Plant; 7: Distributed Energy Storage Plant; 8: Poverty Alleviation Plant; 9: Wind Power Plant; 10: Utility Energy Storage Plant;12: C&I Energy Storage Plant
result_data.pageList.share_type	String	0	Sharing type of the power plant 1: sharing type (browsing rights) 2: sharing type (management rights) 0: non-sharing power plant (my power plant)
result_data.pageList.today_energy	Map	--	Plant Yield Today
result_data.pageList.today_energy.unit	String	kWh	Unit
result_data.pageList.today_energy.value	String	11.77	Numerical Value
result_data.pageList.today_energy_update_time	String	2024-11-14T16:00:50+08:00	Current energy yield Last updated time Format: yyyy-MM-dd'T 'hh :mm:ssXXX
result_data.pageList.today_income	Map	--	Today's revenue of the power plant <value,unit> (when the power plant is an energy storage type, the revenue is energy storage revenue)
result_data.pageList.today_income.unit	String	CNY	Unit
result_data.pageList.today_income.value	String	11.77	Numerical Value
result_data.pageList.today_income_update_time	String	2024-11-14T16:00:50+08:00	Last updated time Format: yyyy-MM-dd'T 'hh :mm:ssXXX
result_data.pageList.total_capcity	Map	--	Total Installed Capacity
result_data.pageList.total_capcity.unit	String	kWp	Unit
result_data.pageList.total_capcity.value	String	330	Numerical Value
result_data.pageList.total_capcity_update_time	String	2022-11-07T15:37:40+08:00	Total installed capacity Last updated time Format:

			yyyy-MM-dd'T 'hh :mm:ssXXX
result_data.pageList.total_energy	Map	--	Plant Total Yield
result_data.pageList.total_energy.unit	String	kWh	Unit
result_data.pageList.total_energy.value	String	11.77	Numerical Value
result_data.pageList.total_energy_update_time	String	2024-11-14T16:00:50+08:00	Last updated time for accumulated energy output of the power plant format: yyyy-MM-dd'T 'hh :mm:ssXXX
result_data.pageList.total_income	Map	--	Cumulative income <value,unit> (When the power plant is energy storage type, the income is energy storage income)
result_data.pageList.total_income.unit	String	CNY	Unit
result_data.pageList.total_income.value	String	11.77	Numerical Value
result_data.pageList.total_income_update_time	String	2024-11-14T16:00:50+08:00	Last update of total revenue
result_data.pageList.valid_flag	Integer	3	Plant Status (1: Normal; 2: Disabled; 3: Connected)
result_data.pageList.province_name	String	AnHui	Province
result_data.pageList.city_name	String	HeFei	City
result_data.pageList.district_name	String	Xicheng District	District
result_data.pageList.grid_connection_status	Integer	1	Grid-connected status 1- Grid-connected 0- Not grid-connected
result_data.pageList.grid_connection_time	Integer	20230303115625	Grid-connected Date
result_data.pageList.month_income_update_time	String	2024-11-14T16:00:50+08:00	Month's Revenue Update Time
result_data.pageList.month_income	Map	--	Month's Revenue (the revenue of storage-type plants is the storage revenue)
result_data.pageList.month_income.unit	String	CNY	Month's Revenue Unit
result_data.pageList.month_income.value	String	11.77	Month's Revenue Value
result_data.rowCount	String	1	Records

2.1 Success Examples

```
{
  "req_serial_num": "20241114f0ff469abf8502b5edb6a80f",
  "result_code": "1",
  "result_msg": "success",
  "result_data": {
    "pageList": [{
      "total_energy": {
        "unit": "度",
        "value": "381.6"
      },
      "alarm_count": 0,
```

```
"latitude": null,
"description": null,
"total_income_update_time": "2024-11-14T16:00:50+08:00",
"valid_flag": 1,
"curr_power": {
  "unit": "W",
  "value": "358"
},
"ps_fault_status": 3,
"district_name": "东城区",
"co2_reduce_update_time": "2024-11-14T16:00:50+08:00",
"city_name": "北京城区",
"install_date": "2024-11-12 18:19:36",
"build_status": 2,
"today_energy_update_time": "2024-11-14T16:00:50+08:00",
"total_energy_update_time": "2024-11-14T16:00:50+08:00",
"ps_type": 4,
"longitude": null,
"total_capcity_update_time": null,
"equivalent_hour_update_time": "2024-11-14T16:00:50+08:00",
"ps_name": "ad优化器572",
"co2_reduce_total": {
  "unit": "千克",
  "value": "269.19"
},
"curr_power_update_time": "2024-11-14T16:00:50+08:00",
"today_income": {
  "unit": "元",
  "value": "95"
},
"grid_connection_status": 0,
"equivalent_hour": {
  "unit": "小时",
  "value": "23.6"
},
"co2_reduce_total_update_time": "2024-11-14T16:00:50+08:00",
"province_name": "北京市",
"ps_location": "123",
"total_income": {
  "unit": "元",
  "value": "270"
},
"total_capcity": {
  "unit": "kWp",
  "value": "4.025"
},
"share_type": "1",
"ps_current_time_zone": "GMT+8",
"today_income_update_time": "2024-11-14T16:00:50+08:00",
"ps_id": 700669565,
"grid_connection_time": null,
"connect_type": 1,
"today_energy": {
  "unit": "度",
```

```
            "value": "95"
        },
        "ps_status": 1,
        "co2_reduce": {
            "unit": "千克",
            "value": "94.715"
        },
        "fault_count": 0
    }],
    "rowCount": 1
  }
}
```

2.2 Failure Example

```
{
  "result_msg": "er_token_login_invalid",
  "result_data": null,
  "result_code": "E00003"
}
```

Query Device List of the Plant

Post /openapi/getDeviceList

Query device list by plant ID.

1. Request Parameters

Request Parameters	Parameter Type	Required	Parameter Example	Parameter Description
ps_id	String	Yes	476178	Plant ID
curPage	Integer	Yes	1	Page
size	Integer	Yes	10	Page Size
is_virtual_unit	String	No	0	Querying Virtual Device? (1: Virtual Device; 0: Physical Device; defaults to 0). Plants, grid connection points, and units are considered virtual devices
device_type_list	List	No	--	Equipment Type List (such as: 11: Plant; 1: Inverter; 3: Grid Connection Point; 17: Unit; defaults to all when empty). Refer to Appendix: Device Type Definitions for details
rel_state	String	No	1	Device Claim Status: (0: Unclaimed; 1: Claimed)
is_get_firmware_version	String	No	0	Get Device Firmware Info? (0: No; 1: Yes; defaults to 0)

```
{
  "curPage":1,
  "size":50,
  "ps_id":689661,
  "device_type_list":[
    1
  ]
}
```

2. Return Parameters

Return Parameters	Data Type	Example Value	Description
req_serial_num	String	20240829c4ba4ec98875ca2f1e7e8175	Request Sequence Number
result_code	String	1	Error Code
result_msg	String	success	Prompt Message
result_data	Map	--	Returned Data
result_data.pageList	List	--	Paginated Data
result_data.pageList.chnnl_id	Integer	4	Device Channel ID
result_data.pageList.communication_dev_sn	String	A2462**0013	Device Communication Device SN
result_data.pageList.dev_fault_status	Integer	4	Device Fault Status: (1: Fault; 2: Alarm; 4: Normal)
result_data.pageList.dev_status	String	1	Device Deployment Status: (0: Undeployed; 1: Deployed)
result_data.pageList.device_code	Integer	247	Device Address Code
result_data.pageList.device_model_code	String	iHomeManager	Device Model Name
result_data.pageList.device_model_id	Integer	367701	Device Model ID
result_data.pageList.device_name	String	Home Energy Manager3	Device Name
result_data.pageList.device_sn	String	A2462**0013	Device SN
result_data.pageList.device_type	Integer	64	Device Type Code
result_data.pageList.factory_name	String	Sunshine Power Co., Ltd	Manufacturer Name
result_data.pageList.ps_id	Integer	473465	Plant ID
result_data.pageList.ps_key	String	7006*6130_64_247_1	Device ps_key (used to query device data)
result_data.pageList.rel_state	Integer	1	Device Claim Status: (0: Unclaimed; 1: Claimed)
result_data.pageList.type_name	String	Home Energy Manager	Device Type Name
result_data.pageList.uuid	Integer	2024**5175	Device UUID
result_data.pageList.firmware_version_info	Map	--	Device Firmware Version
result_data.pageList.firmware_version_info.bat_version	String	SBRBCU-S_22011.01.22	BMS Unit Version No.
result_data.pageList.firmware_version_info.lcd_version	String	SAPPHIRE-H_01011.31.50	Inverter ARM Software Version No.

result_data.pageList.firmware_version_info.mdsp_version	String	SAPPHIRE-H_03011.31.46	MDSP Version
result_data.pageList.firmware_version_info.sdsp_version	String	SUBCTL-S_04011.01.01	SDSP Version
result_data.pageList.firmware_version_info.pvd_version	String	WINET-SV200.001.00.P023	String Monitoring Unit Software Version No.
result_data.pageList.firmware_version_info.cpld_version	String	WINET-SV200.001.00.P023	CPLD Firmware Version No.
result_data.pageList.firmware_version_info.temp_version	String	WINET-SV200.001.00.P023	Temperature board version no.
result_data.pageList.firmware_version_info.m_version	String	WINET-SV200.001.00.P023	Communication Device Software Version No.
result_data.pageList.firmware_version_info.system_version	String	WINET-SV200.001.00.P023	Communication Device Hardware Platform Version No.
result_data.rowCount	Integer	7	Records

2.1 Success Examples


```
{
    "req_serial_num":"202411277d7c4909868d4d9ef4c2ba61",
    "result_code":"1",
    "result_msg":"success",
    "result_data":{
        "pageList":[
            {
                "chnnl_id":5,
                "type_name":"逆变器",
                "ps_key":"689661_1_1_5",
                "device_sn":"A2262065536",
                "dev_status":"1",
                "device_type":1,
                "factory_name":"阳光电源股份有限公司",
                "uuid":3791641,
                "grid_connection_date":"2022-09-27 15:42:08",
                "device_name":"研发楼NB04",
                "dev_fault_status":4,
                "rel_state":1,
                "device_code":1,
                "ps_id":689661,
                "device_model_id":346856,
                "communication_dev_sn":"B2261882811",
                "device_model_code":"SG50CX-P2-CN"
            },
            {
                "chnnl_id":4,
                "type_name":"逆变器",
                "ps_key":"689661_1_1_4",
                "device_sn":"A2271319309",
                "dev_status":"1",
                "device_type":1,
                "factory_name":"阳光电源股份有限公司",
```

```
        "uuid":3785953,
        "grid_connection_date":"2022-09-26 16:57:06",
        "device_name":"主厂房NB01",
        "dev_fault_status":4,
        "rel_state":1,
        "device_code":1,
        "ps_id":689661,
        "device_model_id":346830,
        "communication_dev_sn":"A2262066345",
        "device_model_code":"SG110CX-P2-CN"
    },
    {
        "chnnl_id":3,
        "type_name":"逆变器",
        "ps_key":"689661_1_1_3",
        "device_sn":"A2262065534",
        "dev_status":"1",
        "device_type":1,
        "factory_name":"阳光电源股份有限公司",
        "uuid":3785890,
        "grid_connection_date":"2022-09-26 16:52:06",
        "device_name":"主厂房NB03",
        "dev_fault_status":4,
        "rel_state":1,
        "device_code":1,
        "ps_id":689661,
        "device_model_id":346856,
        "communication_dev_sn":"B2261882806",
        "device_model_code":"SG50CX-P2-CN"
    },
    {
        "chnnl_id":2,
        "type_name":"逆变器",
        "ps_key":"689661_1_1_2",
        "device_sn":"A2271319217",
        "dev_status":"1",
        "device_type":1,
        "factory_name":"阳光电源股份有限公司",
        "uuid":3785846,
        "grid_connection_date":"2022-09-26 16:47:08",
        "device_name":"主厂房NB02",
        "dev_fault_status":4,
        "rel_state":1,
        "device_code":1,
        "ps_id":689661,
        "device_model_id":346830,
        "communication_dev_sn":"A2262066334",
        "device_model_code":"SG110CX-P2-CN"
    },
    {
        "chnnl_id":1,
        "type_name":"逆变器",
        "ps_key":"689661_1_1_1",
        "device_sn":"A2271319326",
```

```
            "dev_status":"1",
            "device_type":1,
            "factory_name":"阳光电源股份有限公司",
            "uuid":3785529,
            "grid_connection_date":"2022-09-26 16:15:30",
            "device_name":"检修楼NB05",
            "dev_fault_status":4,
            "rel_state":1,
            "device_code":1,
            "ps_id":689661,
            "device_model_id":346830,
            "communication_dev_sn":"A2262066326",
            "device_model_code":"SG110CX-P2-CN"
          }
      ],
      "rowCount":5
   }
}
```


2.2 Failure Example


```
{
   "result_msg": "er_token_login_invalid",
   "result_data": null,
   "result_code": "E00003"
}
```


Query Basic Plant Info

Post /openapi/getPowerStationDetail

Query Basic Plant Info

1. Request Parameters

Request Parameters	Parameter Type	Required	Parameter Example	Parameter Description
sn	String	Yes	B2262720904	Device S/N
is_get_ps_remarks	String	Yes	1	Get Plant Remarks?: (1: Yes; defaults to no if not passed)


```
{
   "sn":"B2313140126",
   "is_get_ps_remarks":"1"
}
```


2. Return Parameters

Return Parameters	Data Type	Example Value	Description
req_serial_num	String	202212236d134e3386b3c49b66c966c5	Request Sequence Number

result_code	String	1	Error Code
result_data	Map	--	Returned Data
result_data.alarm_count	Integer	0	Alarms
result_data.build_status	Integer	2	Plant Construction Status (0: Not Started; 1: Under Construction; 2: Connected; 3: Planned; 4: Not Connected)
result_data.communication_dev_detail_list	List	--	Plant Communication Device SN
result_data.communication_dev_detail_list.is_enable	Integer	1	SN Status: (0: Disabled; 1: Normal)
result_data.communication_dev_detail_list.sn	String	A2462**0013	SN
result_data.connect_type	Integer	2	Grid Connection Type: (1: Full Grid; 2: Self-Generated and Self-Used (Surplus Power); 3: Self-Generated and Self-Used (No Feed Network); 4: Off-Grid)
result_data.description	String	Rooftop power station	About the Plant
result_data.design_capacity	Integer	1000000	Installed Power (Wp)
result_data.email	String	155147***64@gmail.com	End User Email
result_data.fault_count	Integer	0	Fault Count
result_data.install_date	String	2022-05-27 11:33:06	Creation Date
result_data.latitude	Num	41.69395123697499	Latitude
result_data.longitude	Num	123.18212151404492	Plant Longitude
result_data.param_income_unit_name	String	CNY	Tariff Unit
result_data.ps_current_time_zone	String	GMT+8	Current Plant Time Zone
result_data.ps_fault_status	Integer	3	Plant Fault Status (1: Fault; 2: Alarm; 3: Normal)
result_data.ps_id	Integer	974621	Plant ID
result_data.ps_key	String	974621_11_0_0	Plant ps_key
result_data.ps_location	String	Longtu Rd, Shushan District, Hefei, Anhui, China	Plant Location
result_data.ps_name	String	PlantA	Project Name
result_data.ps_price	String	0.001	The current electricity price per Wh of the power plant
result_data.ps_price_kwh	String	1	The current electricity price per kWh of the power plant
result_data.ps_status	Integer	0	power plant status 1: online, 0: offline
result_data.ps_type	Integer	3	Plant Type: 1: Utility Plant; 3: Distributed PV Plant; 4: Residential PV Plant; 5: Residential Energy Storage Plant;

			6: Village Plant; 7: Distributed Energy Storage Plant; 8: Poverty Alleviation Plant; 9: Wind Power Plant; 12: C&I Energy Storage Plant
result_data.ps_type_name	String	Commercial PV	Plant Type Name
result_data.share_type	String	2	Sharing type of the power plant 1: sharing type (browsing rights) 2: sharing type (management rights) 0: non-sharing power plant (my power plant)
result_data.share_user_type	String	1	Type of sharing of power plant: 1: owner sharing 2: installer sharing Other: non-sharing power plant
result_data.user_moble_tel	String	185736**246	End User Mobile Number
result_data.ps_remarks	Map	--	Plant Remarks (returned when is_get_ps_remarks is 1)
result_data.ps_remarks.remark1	String	Record rooftop power generation data for the power station	Remark 1
result_data.ps_remarks.remark2	String	Record rooftop power generation data for the power station	Remark 2
result_data.ps_remarks.remark3	String	Record rooftop power generation data for the power station	Remark 3
result_data.ps_remarks.remark4	String	Record rooftop power generation data for the power station	Remark 4
result_data.ps_remarks.remark5	String	Record rooftop power generation data for the power station	Remark 5
result_msg	String	success	Prompt Message

2.1 Success Examples


```
{
    "req_serial_num":"20241127cf2a4725ac6ebd9e9caeab59",
    "result_code":"1",
    "result_msg":"success",
    "result_data":{
        "design_capacity":11000.0,
        "alarm_count":0,
        "ps_key":"974621_11_0_0",
        "latitude":33.07597603786331,
        "description":null,
        "ps_price_kwh":"1",
        "ps_fault_status":3,
        "ps_type_name":"户用光伏",
        "build_status":2,
        "install_date":"2023-06-13 15:40:56",
```

```
        "ps_type":4,
        "email":null,
        "longitude":112.60275055975228,
        "param_income_unit_name":"元",
        "ps_price":"0.001",
        "ps_name":"李庚莲11kwB2313140126",
        "share_user_type":null,
        "ps_location":"河南省南阳市南召县太山庙乡张沟村核西组456号",
        "share_type":"0",
        "ps_current_time_zone":"GMT+8",
        "user_moble_tel":"13037650202",
        "ps_id":974621,
        "communication_dev_detail_list":[
            {
                "is_enable":1,
                "sn":"B2313140126"
            }
        ],
        "connect_type":1,
        "ps_status":1,
        "fault_count":0
    }
}
```

2.2 Failure Example

```
{
    "result_msg": "er_token_login_invalid",
    "result_data": null,
    "result_code": "E00003"
}
```

Query Device Real-Time Measuring Point Data

Post /openapi/getDeviceRealTimeData

Query real-time measuring point data by device ps_key or SN. Note: 1. This endpoint always returns the smallest unit of data. For example, Wh for yield, W for power, A for current, and V for voltage. 2. The data refreshes about every 5 minutes. It is recommended that the endpoint call frequency is no less than 5 minutes.

1. Request Parameters

Request Parameters	Parameter Type	Required	Parameter Example	Parameter Description
point_id_list	List	Yes	["83022","83033"]	Measuring Point ID Set (see Common Measuring Points for specific open measuring point definitions)
ps_key_list	List	No	["972018_11_0_0","971822_11_0_0"]	ps_key Collection (must pass the same device type)
device_type	Integer	Yes	11	Device Type (see Appendix 1 for details)
sn_list	List	No	['A21B1801347','A29B1902347']	SN Collection (pass either the ps_key_list or sn_list)

```
{
    "device_type":11,
    "point_id_list":[
        "83022",
        "83033"
    ],
    "ps_key_list":[
        "972018_11_0_0",
        "971822_11_0_0",
        "972131_11_0_0",
        "972192_11_0_0",
        "972245_11_0_0",
        "971884_11_0_0",
        "971868_11_0_0",
        "972231_11_0_0",
        "971556_11_0_0",
        "971812_11_0_0",
        "971841_11_0_0",
        "972292_11_0_0",
        "971671_11_0_0",
        "972402_11_0_0",
        "972079_11_0_0",
        "972361_11_0_0",
        "971779_11_0_0",
        "971867_11_0_0",
        "972086_11_0_0",
        "972262_11_0_0",
        "971891_11_0_0",
        "972393_11_0_0",
        "972359_11_0_0",
        "972270_11_0_0",
        "972068_11_0_0",
        "972062_11_0_0",
        "972213_11_0_0",
        "971821_11_0_0",
        "972028_11_0_0",
        "972347_11_0_0",
        "971989_11_0_0",
        "971594_11_0_0",
        "971886_11_0_0",
        "972182_11_0_0",
        "972175_11_0_0",
        "971897_11_0_0",
        "971830_11_0_0",
        "971644_11_0_0",
        "971988_11_0_0",
        "972307_11_0_0",
        "972188_11_0_0",
        "971732_11_0_0",
        "972083_11_0_0",
        "972363_11_0_0",
        "972368_11_0_0",
        "972370_11_0_0",
```

```
        "972400_11_0_0",
        "972315_11_0_0",
        "970645_11_0_0",
        "970956_11_0_0"
    ]
}
```

2. Return Parameters

Return Parameters	Data Type	Example Value	Description
req_serial_num	String	2024112728b644398654b9c98150d2f2	Req Seq Num
result_code	String	1	Erro
result_msg	String	success	Pror Mes
result_data	Map	--	Retu
result_data.device_point_list	List	--	Dev Mea Poir
result_data.device_point_list.device_point	Map	--	Poir
result_data.device_point_list.device_point.communication_dev_sn	String	A235**07928	Dev Con Dev
result_data.device_point_list.device_point.dev_fault_status	Integer	4	Dev alar Faul 4: n
result_data.device_point_list.device_point.dev_status	Integer	1	Dev Dep Stat Und Dep
result_data.device_point_list.device_point.device_name	String	Hybrid Inverter2	Dev
result_data.device_point_list.device_point.device_sn	String	A23519**928	Dev
result_data.device_point_list.device_point.device_time	String	20241127151500	Dev Upd
result_data.device_point_list.device_point.p+pointid	Map	--	Mea Poir repr valu poir
result_data.device_point_list.device_point.ps_id	Integer	970645	Plar
result_data.device_point_list.device_point.ps_key	String	970645_11_0_0	Dev
result_data.device_point_list.device_point.uuid	Integer	10941275	Dev
result_data.fail_ps_key_list	List	["700926130_22_247_2","700926130_22_247_3"]	Coll Illeg

2.1 Success Examples

```
{
  "req_serial_num":"2024112728b644398654b9c98150d2f2",
  "result_code":"1",
  "result_msg":"success",
  "result_data":{
    "fail_ps_key_list":[],
    "device_point_list":[
      {
        "device_point":{
          "device_name":"贺伟26.4kWB2342458280的电站陈程程",
          "dev_fault_status":4,
          "ps_key":"970645_11_0_0",
          "device_sn":null,
          "dev_status":1,
          "ps_id":970645,
          "communication_dev_sn":null,
          "uuid":10941275,
          "p83022":"108500.0",
          "p83033":"9276.0",
          "device_time":"20241127151500"
        }
      },
      {
        "device_point":{
          "device_name":"黄拥军26.4kwB2342457345的电站",
          "dev_fault_status":4,
          "ps_key":"970956_11_0_0",
          "device_sn":null,
          "dev_status":1,
          "ps_id":970956,
          "communication_dev_sn":null,
          "uuid":10942796,
          "p83022":"83800.0",
          "p83033":"6290.0",
          "device_time":"20241127151500"
        }
      },
      {
        "device_point":{
          "device_name":"范守勤16.5kwB2320605745",
          "dev_fault_status":4,
          "ps_key":"971556_11_0_0",
          "device_sn":null,
          "dev_status":1,
          "ps_id":971556,
          "communication_dev_sn":null,
          "uuid":10945590,
          "p83022":"26000.0",
          "p83033":"2214.0",
          "device_time":"20241127151500"
        }
      },
      {
```

```
    "device_point":{
      "device_name":"毕家才41.80kwB2332766084的电站",
      "dev_fault_status":4,
      "ps_key":"971594_11_0_0",
      "device_sn":null,
      "dev_status":1,
      "ps_id":971594,
      "communication_dev_sn":null,
      "uuid":10945739,
      "p83022":"67700.0",
      "p83033":"5723.0",
      "device_time":"20241127151500"
    }
  },
  {
    "device_point":{
      "device_name":"杨小二38.15KWB2341295255",
      "dev_fault_status":4,
      "ps_key":"971644_11_0_0",
      "device_sn":null,
      "dev_status":1,
      "ps_id":971644,
      "communication_dev_sn":null,
      "uuid":10945915,
      "p83022":"137800.0",
      "p83033":"14280.0",
      "device_time":"20241127151500"
    }
  },
  {
    "device_point":{
      "device_name":"张菊菊45.1kwB2341876789的电站",
      "dev_fault_status":4,
      "ps_key":"971671_11_0_0",
      "device_sn":null,
      "dev_status":1,
      "ps_id":971671,
      "communication_dev_sn":null,
      "uuid":10946029,
      "p83022":"143300.0",
      "p83033":"14525.0",
      "device_time":"20241127151500"
    }
  },
  {
    "device_point":{
      "device_name":"高兴娃32.155KwB2330772162的电站",
      "dev_fault_status":4,
      "ps_key":"971732_11_0_0",
      "device_sn":null,
      "dev_status":1,
      "ps_id":971732,
      "communication_dev_sn":null,
      "uuid":10946292,
```

```
            "p83022":"81700.0",
            "p83033":"8278.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"刘彦芳22.89KWB22B1862980的电站",
            "dev_fault_status":4,
            "ps_key":"971779_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971779,
            "communication_dev_sn":null,
            "uuid":10946442,
            "p83022":"83200.0",
            "p83033":"8633.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"黄久霞18.15kw82250759121的电站",
            "dev_fault_status":4,
            "ps_key":"971812_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971812,
            "communication_dev_sn":null,
            "uuid":10946607,
            "p83022":"27000.0",
            "p83033":"1756.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"王发胜43.055KwB2340112642",
            "dev_fault_status":4,
            "ps_key":"971821_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971821,
            "communication_dev_sn":null,
            "uuid":10946641,
            "p83022":"99500.0",
            "p83033":"9702.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"张俊奎15.4kwB2342319446的电站",
            "dev_fault_status":4,
```

```
            "ps_key":"971822_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971822,
            "communication_dev_sn":null,
            "uuid":10946646,
            "p83022":"51500.0",
            "p83033":"6752.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"张艳平17.6kwB2342319436的电站",
            "dev_fault_status":4,
            "ps_key":"971830_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971830,
            "communication_dev_sn":null,
            "uuid":10946667,
            "p83022":"23700.0",
            "p83033":"2428.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"吴福地26.16kwB2342458486的电站",
            "dev_fault_status":4,
            "ps_key":"971841_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971841,
            "communication_dev_sn":null,
            "uuid":10946737,
            "p83022":"88000.0",
            "p83033":"7335.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"张惠恩21.8B2330307529",
            "dev_fault_status":4,
            "ps_key":"971867_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971867,
            "communication_dev_sn":null,
            "uuid":10946869,
            "p83022":"54500.0",
            "p83033":"5327.0",
            "device_time":"20241127151500"
```

```
        }
    },
    {
        "device_point":{
            "device_name":"李朝贵 22.55kw B2231810290",
            "dev_fault_status":4,
            "ps_key":"971868_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971868,
            "communication_dev_sn":null,
            "uuid":10946877,
            "p83022":"15400.0",
            "p83033":"1225.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"高建芳29.975kwB2330771095的电站",
            "dev_fault_status":4,
            "ps_key":"971884_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971884,
            "communication_dev_sn":null,
            "uuid":10946972,
            "p83022":"73300.0",
            "p83033":"8559.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"李兴伟 26.4kw B2270844927",
            "dev_fault_status":4,
            "ps_key":"971886_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971886,
            "communication_dev_sn":null,
            "uuid":10946982,
            "p83022":"16400.0",
            "p83033":"1486.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"刘正江20.9kwB2231810464的电站",
            "dev_fault_status":4,
            "ps_key":"971891_11_0_0",
            "device_sn":null,
            "dev_status":1,
```

```
            "ps_id":971891,
            "communication_dev_sn":null,
            "uuid":10947000,
            "p83022":"30600.0",
            "p83033":"2250.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"张学祥 31.35kw B22B1158989",
            "dev_fault_status":4,
            "ps_key":"971897_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971897,
            "communication_dev_sn":null,
            "uuid":10947079,
            "p83022":"65600.0",
            "p83033":"3782.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"郑丁兴42.51kwB2340742192的电站",
            "dev_fault_status":4,
            "ps_key":"971988_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971988,
            "communication_dev_sn":null,
            "uuid":10947534,
            "p83022":"144800.0",
            "p83033":"12724.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"张传侠39.6kwB2341297175的电站",
            "dev_fault_status":4,
            "ps_key":"971989_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":971989,
            "communication_dev_sn":null,
            "uuid":10947543,
            "p83022":"92800.0",
            "p83033":"8614.0",
            "device_time":"20241127151500"
        }
    },
    {
```

```
    "device_point":{
        "device_name":"张凤军12.1kwB2332765440的电站",
        "dev_fault_status":4,
        "ps_key":"972018_11_0_0",
        "device_sn":null,
        "dev_status":1,
        "ps_id":972018,
        "communication_dev_sn":null,
        "uuid":10947672,
        "p83022":"28300.0",
        "p83033":"396.0",
        "device_time":"20241127151500"
    }
},
{
    "device_point":{
        "device_name":"邹银荣24.525kwB2272161584的电站",
        "dev_fault_status":4,
        "ps_key":"972028_11_0_0",
        "device_sn":null,
        "dev_status":1,
        "ps_id":972028,
        "communication_dev_sn":null,
        "uuid":10947700,
        "p83022":"83200.0",
        "p83033":"7806.0",
        "device_time":"20241127151500"
    }
},
{
    "device_point":{
        "device_name":"穆艳峰30.52kwB2350529979的电站",
        "dev_fault_status":4,
        "ps_key":"972062_11_0_0",
        "device_sn":null,
        "dev_status":1,
        "ps_id":972062,
        "communication_dev_sn":null,
        "uuid":10947800,
        "p83022":"70200.0",
        "p83033":"6671.0",
        "device_time":"20241127151500"
    }
},
{
    "device_point":{
        "device_name":"穆艳峰30.52kwB2350530341的电站",
        "dev_fault_status":4,
        "ps_key":"972068_11_0_0",
        "device_sn":null,
        "dev_status":1,
        "ps_id":972068,
        "communication_dev_sn":null,
        "uuid":10947809,
```

```
            "p83022":"70900.0",
            "p83033":"6878.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"刘明会13.2kwB2342072196的电站",
            "dev_fault_status":4,
            "ps_key":"972079_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972079,
            "communication_dev_sn":null,
            "uuid":10947853,
            "p83022":"26900.0",
            "p83033":"405.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"刘子云20.35kwB2271632274的电站",
            "dev_fault_status":4,
            "ps_key":"972083_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972083,
            "communication_dev_sn":null,
            "uuid":10947881,
            "p83022":"46200.0",
            "p83033":"597.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"马双木49.05kwB2341297584的电站",
            "dev_fault_status":4,
            "ps_key":"972086_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972086,
            "communication_dev_sn":null,
            "uuid":10947897,
            "p83022":"169300.0",
            "p83033":"14814.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"辛文书15.805kwB23414E9995的电站",
            "dev_fault_status":4,
```

```
      "ps_key":"972131_11_0_0",
      "device_sn":null,
      "dev_status":1,
      "ps_id":972131,
      "communication_dev_sn":null,
      "uuid":10948096,
      "p83022":"64200.0",
      "p83033":"5021.0",
      "device_time":"20241127151500"
    }
  },
  {
    "device_point":{
      "device_name":"关慧娟30.25kwB23414F0307",
      "dev_fault_status":4,
      "ps_key":"972175_11_0_0",
      "device_sn":null,
      "dev_status":1,
      "ps_id":972175,
      "communication_dev_sn":null,
      "uuid":10948298,
      "p83022":"82200.0",
      "p83033":"7112.0",
      "device_time":"20241127151500"
    }
  },
  {
    "device_point":{
      "device_name":"陆芳39.6kwB2332316940",
      "dev_fault_status":4,
      "ps_key":"972182_11_0_0",
      "device_sn":null,
      "dev_status":1,
      "ps_id":972182,
      "communication_dev_sn":null,
      "uuid":10948320,
      "p83022":"104000.0",
      "p83033":"8781.0",
      "device_time":"20241127151500"
    }
  },
  {
    "device_point":{
      "device_name":"裴吉娜19.62KWB2280420748的电站",
      "dev_fault_status":4,
      "ps_key":"972188_11_0_0",
      "device_sn":null,
      "dev_status":1,
      "ps_id":972188,
      "communication_dev_sn":null,
      "uuid":10948334,
      "p83022":"37200.0",
      "p83033":"3383.0",
      "device_time":"20241127151500"
```

```
        }
    },
    {
        "device_point":{
            "device_name":"肖项杰23.98kwB2262855638的电站",
            "dev_fault_status":4,
            "ps_key":"972192_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972192,
            "communication_dev_sn":null,
            "uuid":10948359,
            "p83022":"0.0",
            "p83033":"0.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"任彪30.25kwB2341873283",
            "dev_fault_status":4,
            "ps_key":"972213_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972213,
            "communication_dev_sn":null,
            "uuid":10948438,
            "p83022":"78400.0",
            "p83033":"6980.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"吴立国19.62kwB22B1986830的电站",
            "dev_fault_status":4,
            "ps_key":"972231_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972231,
            "communication_dev_sn":null,
            "uuid":10948492,
            "p83022":"73100.0",
            "p83033":"5678.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"刘晓双17.985kwB22B1985833的电站",
            "dev_fault_status":4,
            "ps_key":"972245_11_0_0",
            "device_sn":null,
            "dev_status":1,
```

```
            "ps_id":972245,
            "communication_dev_sn":null,
            "uuid":10948540,
            "p83022":"67700.0",
            "p83033":"4111.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"高玉柱20.35kwB2271632174的电站",
            "dev_fault_status":4,
            "ps_key":"972262_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972262,
            "communication_dev_sn":null,
            "uuid":10948628,
            "p83022":"47600.0",
            "p83033":"603.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"王桂荣17.05kwB2330305638的电站",
            "dev_fault_status":4,
            "ps_key":"972270_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972270,
            "communication_dev_sn":null,
            "uuid":10948676,
            "p83022":"31200.0",
            "p83033":"651.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"史变霞22.89kwB2350618723的电站",
            "dev_fault_status":4,
            "ps_key":"972292_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972292,
            "communication_dev_sn":null,
            "uuid":10948795,
            "p83022":"49900.0",
            "p83033":"4778.0",
            "device_time":"20241127151500"
        }
    },
    {
```

```
        "device_point":{
            "device_name":"崔学凤21.255kwB2322379952的电站",
            "dev_fault_status":4,
            "ps_key":"972307_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972307,
            "communication_dev_sn":null,
            "uuid":10948871,
            "p83022":"79100.0",
            "p83033":"7819.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"柳占会33KW B2350425076的电站",
            "dev_fault_status":4,
            "ps_key":"972315_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972315,
            "communication_dev_sn":null,
            "uuid":10948929,
            "p83022":"90000.0",
            "p83033":"8339.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"孙尽松34.335kwB23414F2485的电站",
            "dev_fault_status":4,
            "ps_key":"972347_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972347,
            "communication_dev_sn":null,
            "uuid":10949097,
            "p83022":"42400.0",
            "p83033":"0.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"肖绍全49.05KWB2331584787的电站",
            "dev_fault_status":4,
            "ps_key":"972359_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972359,
            "communication_dev_sn":null,
            "uuid":10949146,
```

```
            "p83022":"26500.0",
            "p83033":"572.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"蒋朝霞32.45kwB2341296122的电站",
            "dev_fault_status":4,
            "ps_key":"972361_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972361,
            "communication_dev_sn":null,
            "uuid":10949161,
            "p83022":"89600.0",
            "p83033":"11702.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"商桂敏49.05KWB231586521的电站",
            "dev_fault_status":4,
            "ps_key":"972363_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972363,
            "communication_dev_sn":null,
            "uuid":10949165,
            "p83022":"12300.0",
            "p83033":"341.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"张春刚13.08kwB2270844640的电站",
            "dev_fault_status":4,
            "ps_key":"972368_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972368,
            "communication_dev_sn":null,
            "uuid":10949177,
            "p83022":"42400.0",
            "p83033":"3439.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"吴志军49.05KWB2332318681的电站",
            "dev_fault_status":4,
```

```
            "ps_key":"972370_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972370,
            "communication_dev_sn":null,
            "uuid":10949188,
            "p83022":"3100.0",
            "p83033":"0.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"陆晓艳34.10kwB2350426997的电站",
            "dev_fault_status":4,
            "ps_key":"972393_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972393,
            "communication_dev_sn":null,
            "uuid":10949277,
            "p83022":"116200.0",
            "p83033":"5510.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"乔艳飞38.15kwB2350530797的电站",
            "dev_fault_status":4,
            "ps_key":"972400_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972400,
            "communication_dev_sn":null,
            "uuid":10949310,
            "p83022":"0.0",
            "p83033":"0.0",
            "device_time":"20241127151500"
        }
    },
    {
        "device_point":{
            "device_name":"乔艳飞46.325kwB2250904540的电站",
            "dev_fault_status":4,
            "ps_key":"972402_11_0_0",
            "device_sn":null,
            "dev_status":1,
            "ps_id":972402,
            "communication_dev_sn":null,
            "uuid":10949318,
            "p83022":"0.0",
            "p83033":"0.0",
            "device_time":"20241127151500"
```

```
            }
        }
    ]
  }
}
```

2.2 Failure Example


```
{
  "result_msg": "er_token_login_invalid",
  "result_data": null,
  "result_code": "E00003"
}
```


Query Historical Minute-Level Measuring Point Data of Devices of the Same Type

Post /openapi/getDevicePointMinuteDataList

Query the minute-level data of device measuring points based on the ps_key and measuring point ID of devices of the same type and the specified time interval (time zone where the plant is located). Supports time intervals of up to 3 hours and up to 50 devices. The endpoint will not return measuring points with no data. Supports up to 50 measuring points per request. Historical data endpoints do not return data for the current day.

1. Request Parameters

Request Parameters	Parameter Type	Required	Parameter Example	Parameter Description
end_time_stamp	String	Yes	20211122080000	End Time (time zone where the plant is located; format: yyyyMMddHHmmss)
minute_interval	String	No	15	Duration Interval in Minutes (15 means measuring point data is queried every 15 minutes; if the parameter is not passed or empty, measuring point data is queried every 5 minutes)
points	String	Yes	p83024	p+ Measuring Point ID (such as p3022,p3024, separate multiple measuring points with commas; see Common Measuring Points for specific open measuring point definitions)
ps_key_list	List	Yes	["184741_1_1_1"]	Device ps_key Collection (devices of the same type)
start_time_stamp	String	Yes	20240724110000	Start Time (time zone where the plant is located; format: yyyyMMddHHmmss)
is_get_data_acquisition_time	String	No	1	Return Measuring Point Dictionary? (0: No; 1: Yes; defaults to no)

```
{
  "start_time_stamp":"20211122080000",
  "points":"p5,p6,p7,p8,p9,p10,p18,p19,p20,p21,p22,p23,p24,p45,p46,p47,p48,p49,p50,p51,p52,p53,p54",
  "end_time_stamp":"20211122105000",
  "minute_interval":10,
  "ps_key_list":[
    "184741_1_1_1"
  ],
```

```
    "is_get_data_acquisition_time": "1"
}
```

2. Return Parameters

Return Parameters	Data Type	Example Value	Description
req_serial_num	String	202411275316448b87dbc526d8a5efc4	Request Sequence Number
result_code	String	1	Error Code
result_msg	String	success	Prompt Message
result_data.{ps_key}.{p+pointid}	String	p23	Measuring Point Value (output parameters are as follows: p1, p83022)
result_data	Map	--	Returned Data
result_data.{ps_key}	List	["700926130_22_247_2","700926130_22_247_3"]	Device ps_key
result_data.{ps_key}.time_stamp	String	20211122080000	Time (time zone where the plant is located)
result_data.{ps_key}.pST001	String	20211122105000	Data Upload Time

2.1 Success Examples

```
{
    "req_serial_num":"202411275316448b87dbc526d8a5efc4",
    "result_code":"1",
    "result_msg":"success",
    "result_data":{
        "184741_1_1_1":[
            {
                "p23":"1.1",
                "p22":"1.0",
                "p24":"626.0",
                "time_stamp":"20211122080000",
                "p5":"474.6",
                "p6":"0.5",
                "p18":"230.6",
                "p7":"673.3",
                "p8":"0.5",
                "p19":"229.3",
                "p21":"1.2",
                "p20":"229.3"
            },
            {
                "p23":"1.2",
                "p22":"1.2",
                "p24":"708.0",
                "time_stamp":"20211122081000",
                "p5":"492.8",
                "p6":"0.6",
                "p18":"230.4",
                "p7":"680.9",
                "p8":"0.6",
```

```
        "p19":"229.4",
        "p21":"1.2",
        "p20":"230.1"
    },
    {
        "p23":"2.0",
        "p22":"2.0",
        "p24":"1332.0",
        "time_stamp":"20211122082000",
        "p5":"507.0",
        "p6":"1.2",
        "p18":"230.0",
        "p7":"701.0",
        "p8":"1.1",
        "p19":"229.0",
        "p21":"2.0",
        "p20":"229.3"
    },
    {
        "p23":"2.6",
        "p22":"2.6",
        "p24":"1757.0",
        "time_stamp":"20211122083000",
        "p5":"522.5",
        "p6":"1.5",
        "p18":"230.0",
        "p7":"718.6",
        "p8":"1.4",
        "p19":"229.0",
        "p21":"2.6",
        "p20":"228.9"
    },
    {
        "p23":"2.7",
        "p22":"2.8",
        "p24":"1848.0",
        "time_stamp":"20211122084000",
        "p5":"526.6",
        "p6":"1.5",
        "p18":"231.0",
        "p7":"723.2",
        "p8":"1.5",
        "p19":"229.8",
        "p21":"2.7",
        "p20":"228.8"
    },
    {
        "p23":"2.7",
        "p22":"2.7",
        "p24":"1809.0",
        "time_stamp":"20211122085000",
        "p5":"521.5",
        "p6":"1.5",
        "p18":"230.3",
```

```
        "p7":"720.7",
        "p8":"1.5",
        "p19":"230.0",
        "p21":"2.6",
        "p20":"229.2"
    },
    {
        "p23":"2.0",
        "p22":"1.9",
        "p24":"1301.0",
        "time_stamp":"20211122090000",
        "p5":"508.3",
        "p6":"1.1",
        "p18":"230.5",
        "p7":"707.1",
        "p8":"1.1",
        "p19":"228.6",
        "p21":"1.9",
        "p20":"228.7"
    },
    {
        "p23":"1.9",
        "p22":"2.0",
        "p24":"1256.0",
        "time_stamp":"20211122091000",
        "p5":"505.0",
        "p6":"1.1",
        "p18":"229.3",
        "p7":"710.4",
        "p8":"1.0",
        "p19":"228.6",
        "p21":"1.9",
        "p20":"228.1"
    },
    {
        "p23":"1.7",
        "p22":"1.7",
        "p24":"1068.0",
        "time_stamp":"20211122092000",
        "p5":"500.1",
        "p6":"1.0",
        "p18":"229.1",
        "p7":"691.8",
        "p8":"0.9",
        "p19":"229.0",
        "p21":"1.6",
        "p20":"228.4"
    },
    {
        "p23":"2.0",
        "p22":"2.1",
        "p24":"1355.0",
        "time_stamp":"20211122093000",
        "p5":"506.9",
```

```
        "p6":"1.1",
        "p18":"229.8",
        "p7":"703.1",
        "p8":"1.1",
        "p19":"229.3",
        "p21":"2.1",
        "p20":"228.9"
    },
    {
        "p23":"2.7",
        "p22":"2.7",
        "p24":"1829.0",
        "time_stamp":"20211122094000",
        "p5":"524.0",
        "p6":"1.5",
        "p18":"230.5",
        "p7":"729.4",
        "p8":"1.5",
        "p19":"229.3",
        "p21":"2.7",
        "p20":"229.3"
    },
    {
        "p23":"3.2",
        "p22":"3.3",
        "p24":"2228.0",
        "time_stamp":"20211122095000",
        "p5":"520.0",
        "p6":"1.9",
        "p18":"231.8",
        "p7":"724.2",
        "p8":"1.8",
        "p19":"230.3",
        "p21":"3.2",
        "p20":"230.3"
    },
    {
        "p23":"3.4",
        "p22":"3.4",
        "p24":"2305.0",
        "time_stamp":"20211122100000",
        "p5":"524.5",
        "p6":"1.9",
        "p18":"231.6",
        "p7":"733.1",
        "p8":"1.9",
        "p19":"230.4",
        "p21":"3.4",
        "p20":"230.5"
    },
    {
        "p23":"5.5",
        "p22":"5.4",
        "p24":"3775.0",
```

```
        "time_stamp":"20211122101000",
        "p5":"536.5",
        "p6":"3.0",
        "p18":"231.9",
        "p7":"743.7",
        "p8":"3.0",
        "p19":"230.7",
        "p21":"5.4",
        "p20":"230.8"
    },
    {
        "p23":"4.3",
        "p22":"4.3",
        "p24":"2974.0",
        "time_stamp":"20211122102000",
        "p5":"530.2",
        "p6":"2.4",
        "p18":"231.3",
        "p7":"736.8",
        "p8":"2.4",
        "p19":"230.5",
        "p21":"4.3",
        "p20":"229.4"
    },
    {
        "p23":"4.6",
        "p22":"4.6",
        "p24":"3161.0",
        "time_stamp":"20211122103000",
        "p5":"529.2",
        "p6":"2.5",
        "p18":"230.9",
        "p7":"738.1",
        "p8":"2.5",
        "p19":"229.6",
        "p21":"4.6",
        "p20":"229.9"
    },
    {
        "p23":"5.3",
        "p22":"5.3",
        "p24":"3642.0",
        "time_stamp":"20211122104000",
        "p5":"537.3",
        "p6":"2.9",
        "p18":"231.1",
        "p7":"745.7",
        "p8":"2.9",
        "p19":"229.1",
        "p21":"5.2",
        "p20":"229.2"
    },
    {
        "p23":"5.0",
```

```
            "p22":"5.0",
            "p24":"3451.0",
            "time_stamp":"20211122105000",
            "p5":"532.1",
            "p6":"2.7",
            "p18":"231.2",
            "p7":"742.3",
            "p8":"2.7",
            "p19":"228.6",
            "p21":"5.0",
            "p20":"229.4"
          }
        ]
      }
  }
```

2.2 Failure Example

```
{
   "result_msg": "er_token_login_invalid",
   "result_data": null,
   "result_code": "E00003"
}
```

3. More Information

3.1 Device Measuring Points

Device Type	Measuring Point ID	Measuring Point Name	Unit
Inverter	1	Yield Today	Wh
	87	Yield This Month	Wh
	88	Yield This Year	Wh
	2	Total Yield	Wh
	67	Daily Yield (Theoretical)	Wh
	24	Total Active Power	W
	25	Total Reactive Power	var
	14	Total DC Power	W
	43	Total Apparent Power	VA
	26	Total Power Factor	
	18	Phase A Voltage	V
	19	Phase B Voltage	V
	20	Phase C Voltage	V
	21	Phase A Current	A
	22	Phase B Current	A
	23	Phase C Current	A
	94	Array Insulation Resistahce	kΩ
	27	Grid Frequency	Hz

	7356	Total Operation Time	h
	3	Total On-grid Running Time	h
	15	A-B Line Voltage	V
	16	B-C Line Voltage	V
	17	C-A Line Voltage	V
	95	Bus Voltage	V
	90	Negative Voltage to Ground	V
	120	Number of AFCI Faults	
	4	Internal Air Temperature	°C
	33	Module 1 Temperature	°C
	34	Module 2 Temperature	°C
	35	Module 3 Temperature	°C
	36	Module 4 Temperature	°C
	37	Module 5 Temperature	°C
	38	Module 6 Temperature	°C
	39	Positive Impedance to Ground	Ω
	40	Negative Impedance to Ground	Ω
	41	Active Power Regulation Actual Value	W
	42	Reactive Power Regulation Actual Value	va
	5	MPPT1 Voltage	V
	7	MPPT2 Voltage	V
	9	MPPT3 Voltage	V
	45	MPPT4 Voltage	V
	47	MPPT5 Voltage	V
	49	MPPT6 Voltage	V
	51	MPPT7 Voltage	V
	53	MPPT8 Voltage	V
	55	MPPT9 Voltage	V
	57	MPPT10 Voltage	V
	7401	MPPT11 Voltage	V
	7402	MPPT12 Voltage	V
	7723	MPPT13 Voltage	V
	7725	MPPT14 Voltage	V
	7727	MPPT15 Voltage	V
	7729	MPPT16 Voltage	V
	7731	MPPT17 Voltage	V
	7733	MPPT18 Voltage	V
	7735	MPPT19 Voltage	V
	7737	MPPT20 Voltage	V
	6	MPPT1 Current	A
	8	MPPT2 Current	A
	10	MPPT3 Current	A
	46	MPPT4 Current	A
	48	MPPT5 Current	A

	50	MPPT6 Current	A
	52	MPPT7 Current	A
	54	MPPT8 Current	A
	56	MPPT9 Current	A
	58	MPPT10 Current	A
	7451	MPPT11 Current	A
	7452	MPPT12 Current	A
	7724	MPPT13 Current	A
	7726	MPPT14 Current	A
	7728	MPPT15 Current	A
	7730	MPPT16 Current	A
	7732	MPPT17 Current	A
	7734	MPPT18 Current	A
	7736	MPPT19 Current	A
	7738	MPPT20 Current	A
	96	String 1 Voltage	V
	97	String 2 Voltage	V
	98	String 3 Voltage	V
	99	String 4 Voltage	V
	100	String 5 Voltage	V
	101	String 6 Voltage	V
	102	String 7 Voltage	V
	103	String 8 Voltage	V
	104	String 9 Voltage	V
	105	String 10 Voltage	V
	106	String 11 Voltage	V
	107	String 12 Voltage	V
	108	String 13 Voltage	V
	109	String 14 Voltage	V
	110	String 15 Voltage	V
	111	String 16 Voltage	V
	112	String 17 Voltage	V
	113	String 18 Voltage	V
	7166	String 19 Voltage	V
	7167	String 20 Voltage	V
	7168	String 21 Voltage	V
	7169	String 22 Voltage	V
	7170	String 23 Voltage	V
	7171	String 24 Voltage	V
	7172	String 25 Voltage	V
	7173	String 26 Voltage	V
	7174	String 27 Voltage	V
	7175	String 28 Voltage	V
	7176	String 29 Voltage	V

	7177	String 30 Voltage	V
	7178	String 31 Voltage	V
	7179	String 32 Voltage	V
	7707	String 33 Voltage	V
	7709	String 34 Voltage	V
	7711	String 35 Voltage	V
	7713	String 36 Voltage	V
	7715	String 37 Voltage	V
	7717	String 38 Voltage	V
	7719	String 39 Voltage	V
	7721	String 40 Voltage	V
	70	String 1 Current	A
	71	String 2 Current	A
	72	String 3 Current	A
	73	String 4 Current	A
	74	String 5 Current	A
	75	String 6 Current	A
	76	String 7 Current	A
	77	String 8 Current	A
	78	String 9 Current	A
	79	String 10 Current	A
	80	String 11 Current	A
	81	String 12 Current	A
	82	String 13 Current	A
	83	String 14 Current	A
	84	String 15 Current	A
	85	String 16 Current	A
	92	String 17 Current	A
	93	String 18 Current	A
	313	String 19 Current	A
	314	String 20 Current	A
	315	String 21 Current	A
	316	String 22 Current	A
	317	String 23 Current	A
	318	String 24 Current	A
	319	String 25 Current	A
	320	String 26 Current	A
	321	String 27 Current	A
	322	String 28 Current	A
	323	String 29 Current	A
	324	String 30 Current	A
	325	String 31 Current	A
	326	String 32 Current	A
	7708	String 33 Current	A

	7710	String 34 Current	A
	7712	String 35 Current	A
	7714	String 36 Current	A
	7716	String 37 Current	A
	7718	String 38 Current	A
	7720	String 39 Current	A
	7722	String 40 Current	A
Combiner Box	1002	Interior Temperature	°C
	1001	DC Bus Voltage	V
	1006	Total DC Power	W
	1005	Total DC Current	A
	1009	Ipv-1	A
	1010	Ipv-2	A
	1011	Ipv-3	A
	1012	Ipv-4	A
	1013	Ipv-5	A
	1014	Ipv-6	A
	1015	Ipv-7	A
	1016	Ipv-8	A
	1017	Ipv-9	A
	1018	Ipv-10	A
	1019	Ipv-11	A
	1020	Ipv-12	A
	1021	Ipv-13	A
	1022	Ipv-14	A
	1023	Ipv-15	A
	1024	Ipv-16	A
	1025	Ipv-17	A
	1026	Ipv-18	A
	1027	Ipv-19	A
	1028	Ipv-20	A
	1029	Ipv-21	A
	1030	Ipv-22	A
	1031	Ipv-23	A
	1032	Ipv-24	A
Meteo Station	2003	Transient Horizontal Irradiation	W/㎡
	2001	Daily Horizontal Irradiation	Wh/㎡
	2002	Total Horizontal Irradiation	Wh/㎡
	2007	Slope Transient Irradiation	W/㎡
	2005	Slope Daily Irradiation	Wh/㎡
	2006	Total Slope Irradiation	Wh/㎡
	2009	Ambient Temperature	°C
	2010	Temp. (PV module)	°C
	2016	Wind Speed	m/s

	2011	Wind Speed Scale	
	2012	Wind Angle	°
	2014	Atmospheric Pressure	hPa
	2015	Ambient Humidity	%RH
	2022	Rainfall	mm
	2026	Precipitation	mm
	2100	Temperature 1 (Backplane Temperature)	°C
	2101	Temperature 2	°C
	2102	Temperature 3	°C
	2103	Temperature 4	°C
	2104	Temperature 5	°C
	2105	Dew Point	
	2106	Soil Moisture 1	
	2107	Soil Moisture 2	
	2108	Soil Moisture 3	
	2109	CO_2	
	2110	Evaporation	
	2111	Instantaneous Value of Total Radiation 1	
	2114	Instantaneous Value of Total Radiation 2	
	2112	Instantaneous Value of Scattering Radiation	
	2113	Instantaneous Value of Direct Radiation	
	2115	Instantaneous Value of Net Radiation	
	2116	Instantaneous Value of Photosynthetic Radiation	
	2117	Instantaneous Value of Ultraviolet Radiation	
	2118	Instantaneous Value of Wind Direction	
	2119	Instantaneous Value of Wind Speed	
	2120	2-Minute Wind Speed	
	2121	10-Minute Wind Speed	
	2122	Cumulative Value of Rainfall Time Interval	
	2123	Cumulative Value of Sunshine Time Interval	
	2124	Cumulative Value of Total Radiation 1 Time Interval	
	2127	Cumulative Value of Total Radiation 2 Time Interval	
	2125	Cumulative Value of Scattering Radiation Time Interval	
	2126	Cumulative Value of Direct Radiation Time Interval	
	2128	Cumulative Value of Net Radiation Time Interval	
	2129	Cumulative Value of Photosynthetic Radiation Time Interval	
	2130	Cumulative Value of Ultraviolet Radiation Time Interval	
	2131	Daliy Accumulation of Rainfall	
	2132	Daliy Accumulation of Sunshine	
	2133	Daily Cumulative Value of Total Radiation 1	
	2136	Daily Cumulative Value of Total Radiation 1	
	2134	Daily Cumulative Value of Scattering Radiation	
	2135	Daily Cumulative Value of Direct Radiation	
	2137	Daily Cumulative Value of Net Radiation	

	2138	Daily Cumulative Value of Photosynthetic Radiation	
	2139	Daily Cumulative Value of Ultraviolet Radiation	
Meter	8030	Forward Active Energy	Wh
	8031	Reverse Active Energy	Wh
	8062	Daily Forward Active Energy	Wh
	8063	Daily Reverse Active Energy	Wh
	8032	Forward Reactive Energy	varh
	8033	Reverse Reactive Energy	varh
	8034	Peak Forward Active Energy	Wh
	8035	Peak Reverse Active Energy	Wh
	8038	Valley Forward Active Energy	Wh
	8039	Valley Reverse Active Energy	Wh
	8042	Flat Forward Active Energy	Wh
	8043	Flat Reverse Active Energy	Wh
	8058	Wave Forward Active Energy	Wh
	8059	Wave Reverse Active Energy	Wh
	8036	Peak Forward Reactive Energy	Wh
	8037	Peak Reverse Reactive Energy	Wh
	8040	Valley Forward Reactive Energy	Wh
	8041	Valley Reverse Reactive Energy	Wh
	8044	Flat Forward Reactive Energy	Wh
	8045	Flat Reverse Reactive Energy	Wh
	8060	Wave Forward Reactive Energy	Wh
	8061	Wave Reverse Reactive Energy	Wh
	8000	Phase A Voltage	V
	8001	Phase B Voltage	V
	8002	Phase C Voltage	V
	8003	A-B Line Voltage	V
	8004	B-C Line Voltage	V
	8005	C-A Line Voltage	V
	8006	Phase A Current	A
	8007	Phase B Current	A
	8008	Phase C Current	A
	8064	Frequency	Hz
	8018	Meter Active Power	W
	8022	Reactive Power	var
	8014	PF	
	8026	Apparent Power	VA
	8076	Meter Phase A Active Power	W
	8077	Meter Phase B Active Power	W
	8078	Meter Phase C Active Power	W
	8084	Daily Direct Power Consumption	Wh
	8085	Total Direct Power Consumption	Wh
Communication	10555	AI Voltage Signal 1	V

	10557	AI Voltage Signal 2	V
	10559	AI Voltage Signal 3	V
	10561	AI Voltage Signal 4	V
	10575	AI Current Signal 1	mA
	10576	AI Current Signal 2	mA
	10577	AI Current Signal 3	mA
	10578	AI Current Signal 3	mA
	10563	PT signal 1	°C
	10565	PT signal 2	°C
	10567	DO signal 1	
	10569	DO signal 2	
	10571	DO signal 3	
	10573	DO signal 4	
	10026	Total Power	W
	10046	Total Reactive Power	var
	10587	Total Active Power	W
	10510	Mobile Network Signal Strength	
	10511	WLAN Signal Strength	
	10028	Total Yield	Wh
Energy Storage System	13011	Active Power	W
	13012	Total Reactive Power	var
	13003	Total DC Power	W
	13013	Total Power Factor	
	13157	Phase A Voltage	V
	13158	Phase B Voltage	V
	13159	Phase C Voltage	V
	13008	Phase A Current	A
	13009	Phase B Current	A
	13010	Phase C Current	A
	13160	Array Insulation Resistance	kΩ
	13007	Grid Frequency	Hz
	18065	Phase A Backup Power	W
	18066	Phase B Backup Power	W
	18067	Phase C Backup Power	W
	18068	Total Backup Power	W
	18062	Phase A Backup Current	A
	18063	Phase B Backup Current	A
	18064	Phase C Backup Current	A
	13020	Total Operation Time	H
	13112	Daily PV Yield	Wh
	13134	Total PV Yield	Wh
	13187	AC Voltage	V
	13188	AC Current	A

	13004	A-B Line Voltage	V
	13005	B-C Line Voltage	V
	13006	C-A Line Voltage	V
	13019	Internal Air Temperature	°C
	13161	Bus Voltage	V
	13001	MPPT1 Voltage	V
	13105	MPPT2 Voltage	V
	13107	MPPT3 Voltage	V
	13109	MPPT4 Voltage	V
	13002	MPPT1 Current	A
	13106	MPPT2 Current	A
	13108	MPPT3 Current	A
	13110	MPPT4 Current	A
	13122	Feed-in Energy Today	Wh
	13125	Total Feed-in Energy	Wh
	13147	Energy Purchased Today	Wh
	13148	Total Purchased Energy	Wh
	13149	Purchased Power	W
	13121	Feed-in Power	W
	13173	Feed-in Energy Today (PV)	Wh
	13175	Total Feed-in Energy (PV)	Wh
	13141	Battery Level (SOC)	
	13029	Battery Discharging Energy Today	Wh
	13028	Battery Charging Energy Today	Wh
	13138	Battery Voltage	V
	13139	Battery Current	A
	13035	Total Battery Discharging Energy	Wh
	13034	Total Battery Charging Energy	Wh
	13142	Battery Health (SOH)	
	13143	Battery Temperature	°C
	13162	Max. Charging Current (BMS)	A
	13163	Max. Discharging Current (BMS)	A
	13174	Daily Battery Charging Energy from PV	Wh
	13176	Total Battery Charging Energy from PV	Wh
	13126	Battery Charging Power	W
	13150	Battery Discharging Power	W
	13199	Daily Load Consumption	Wh
	13137	Total Load Energy Consumption from PV	Wh
	13119	Load Power	W
	13130	Total Load Consumption	Wh
	13116	Daily Direct Energy Consumption	Wh
	13144	Daily Self-consumption Rate	
	13016	Total Charging Time	H
	13017	Total Discharging Time	H

	13018	Total Apparent Power	VA
	13023	Daily Charging Time	H
	13024	Daily Discharging Time	H
	13118	Annual Direct Energy Consumption	Wh
	13165	MDSP Off-Grid Start Up Status	
	13166	SDSP Working Mode	
	13167	SDSP Off-grid Start Status	
	13168	DI Status	
	13169	Battery Voltage (BMS)	V
	13170	Battery SOC (BMS)	
	13171	EMS Status	
	13172	Daily Self-Sufficiency Rate	
	13140	Battery Capacity (kWh)	Wh
	18075	Channel 2 Total Apparent Power	VA
	18076	Channel 2 Phase A Apparent Power	VA
	18077	Channel 2 Phase B Apparent Power	VA
	18078	Channel 2 Phase C Apparent Power	VA
	18079	Channel 2 Total Active Power	W
	18080	Channel 2 Phase A Active Power	W
	18081	Channel 2 Phase B Active Power	W
	18082	Channel 2 Phase C Active Power	W
	18083	Channel 2 Total Reactive Power	var
	18084	Channel 2 Phase A Reactive Power	var
	18085	Channel 2 Phase B Reactive Power	var
	18086	Channel 2 Phase C Reactive Power	var
	18087	Channel 2 Power Factor	
	18088	Channel 2 Total Purchased Energy	Wh
	18089	Channel 2 Phase A Purchased Energy	Wh
	18090	Channel 2 Phase B Purchased Energy	Wh
	18091	Channel 2 Phase C Purchased Energy	Wh
	18092	Channel 2 Total Feed-in Energy	Wh
	18093	Channel 2 Phase A Feed-in Energy	Wh
	18094	Channel 2 Phase B Feed-in Energy	Wh
	18095	Channel 2 Phase C Feed-in Energy	Wh
	18103	Phase A Backup Voltage	V
	18104	Phase B Backup Voltage	V
	18105	Phase C Backup Voltage	V
	18108	Meter Phase-A Voltage	V
	18109	Meter Phase-B Voltage	V
	18110	Meter Phase-C Voltage	V
Communication Module	23001	Wireless Signal Strength	
	23014	WLAN Signal Strength	
	23008	Card ID	

Battery	58601	Battery Voltage	V
	58602	Battery Current	A
	58603	Battery Temperature	°C
	58604	Battery Level	
	58605	Battery Health (SOH)	
	58606	Total Battery Charging Energy	Wh
	58607	Total Battery Discharging Energy	Wh
	58608	Battery Operation Status(0: Unplugged, 1: Standby, 2: Run, 3: Fault, 4: Shutdown, 5: Upgrade, 6: Correction, 7: Test)	
	58609	Standard Health Status	
	58610	Max.Voltage of Cell	mV
	58611	Position of Max-Voltage Cell	
	58612	Min.Voltage of Cell	mV
	58613	Position of Min-Voltage Cell	
	58614	Max.Temperature of Module	°C
	58615	Max.Temperature Position of Module	
	58616	Min.Temperature of Module	°C
	58617	Min.Temperature Position of Module	
	58618	Max. Cell Voltage of Module 1	mV
	58619	Max. Cell Voltage of Module 2	mV
	58620	Max. Cell Voltage of Module 3	mV
	58621	Max. Cell Voltage of Module 4	mV
	58622	Max. Cell Voltage of Module 5	mV
	58623	Max. Cell Voltage of Module 6	mV
	58624	Max. Cell Voltage of Module 7	mV
	58625	Max. Cell Voltage of Module 8	mV
	58626	Min. Cell Voltage of Module 1	mV
	58627	Min. Cell Voltage of Module 2	mV
	58628	Min. Cell Voltage of Module 3	mV
	58629	Min. Cell Voltage of Module 4	mV
	58630	Min. Cell Voltage of Module 5	mV
	58631	Min. Cell Voltage of Module 6	mV
	58632	Min. Cell Voltage of Module 7	mV
	58633	Min. Cell Voltage of Module 8	mV
	58635	DC Contactor Status	
	58636	Fault Module ID	
EMS	24620	ESS Daily Charge	Wh
	24622	ESS Total Charge	Wh
	24621	ESS Daily Discharge	Wh
	24623	ESS Total Discharge	Wh
	24624	PV Active Power	W
	24625	Energy Storage Active Power	W
	24626	Grid Active Power	W
	24627	Daily PV Yield	Wh

	24628	Total PV Yield	Wh
	24629	Energy Storage SOC	%
	24630	Energy Storage Remaining Charge	Wh
	24631	Active Load	W
LC	59502	Software Version	
	59546	Daily Charge	Wh
	59535	Total Charge	Wh
	59547	Daily Discharge	Wh
	59536	Total Discharge	Wh
	59541	Total Active Power	W
	59542	Total Reactive Power	var
	59551	Working Mode	
	59552	Working Status (0: Initial Status, 1: Self-test, 2: Self-test Failure, 3: Starting Up, 4: Running, 5: Warn Run, 6: Standby, 7: Stopped, 8: Shutdown, 9: Emergency Stop, 10: Shut Down due to Faults)	
	59705	Device S/N	
PCS	44010	Overall Total Active Power	W
	44011	Overall total reactive power	var
	44012	Overall PF	
	44029	Overall Grid Frequency	Hz
	44025	Work Mode	
	44231	Running Status (0: On-grid Operation, 4096: Backup Operation, 4608: Initial Standby, 5632: Starting Up, 37120: Warn Run, 33024: Derating Running, 5120: Hot Standby, 4864: Press to Shut Down, 21760: Shut Down due to Faults, 5376: Emergency Stop, 12288: Running with Zero Power, 4610: Running)	
	44244	Internal Air Temperature	°C
	44803	Device S/N	
	44823	PCS-DSP Version Number	
	44824	PCS-CPLD Version Number	
CMU	59002	total voltage	
	59003	Total Current	
	59004	SOC	
	59005	SOH	
	59006	rack cap	
	59008	Highest cell voltage	mV
	59010	Lowest Cell Voltage	mV
	59012	Highest cell temperature	°C
	59014	Min Cell Temperature	°C
	59403	Software Version	
BSC	59008	Highest cell voltage	mV
	59010	Lowest Cell Voltage	mV
	59012	Highest cell temperature	°C
	59014	Min Cell Temperature	°C
	59064	Max Cell Voltage Position (#BMU~#Cell)	
	59065	Min Cell Voltage Position (#BMU~#Cell)	
	59066	Max Cell Temperature Position (BMU~#Cell)	

	59067	Min Cell Temperature Position (BMU~#Cell)	
Charger	33708	Charging Power	kW
	33722	Min. Charge Power	kW
	33723	Max. Charging Power	kW
	33702	Phase A Charging Voltage	V
	33704	Phase B Charging Voltage	V
	33706	Phase C Charging Voltage	V
	33710	CP Voltage	V
	33707	Phase C Charging Current	A
	33705	Phase B Charging Current	A
	33703	Phase A Charging Current	A
	33728	Max. Charging Current	A
	33729	Min. Charge Current	A
	33716	Charging Status (1: Idle (not plugged in)--2: Standby (plugged in)--3: Charging--4: Charging paused (station)--5: Charging paused (vehicle)--6: Charging completed--7: Reserved--8: Disabled--9: Fault)	
Micro Inverter	51301	Running Status (21760: Shut Down due to Faults, 0: On-grid Operation, 1024: Maintenance Mode Operation, 2048: forced mode operation, 4096: Backup Operation, 8192: Open loop, 4608: Initial Standby, 32768: sShutdown, 5120: Standby, 5632: Starting Up, 33024: Derating Running, 33280: Dispatch Running, 4369: Initial Status, 4864: Press to Shut Down, 37120: Warn Run)	
	51346	Yield Today	Wh
	51302	Total Yield	Wh
	51303	Total Active Power	W
	51304	Total Reactive Power	var
	51305	Total DC Power	W
	51306	Total Apparent Power	VA
	51307	Total Power Factor	
	51308	Grid Frequency	Hz
	51309	AC Voltage	V
	51312	AC Current	A
	51315	PV1 Voltage	V
	51317	PV2 Voltage	V
	51319	PV3 Voltage	V
	51321	PV4 Voltage	V
	51323	PV5 Voltage	V
	51325	PV6 Voltage	V
	51316	PV1 Current	A
	51318	PV2 Current	A
	51320	PV3 Current	A
	51322	PV4 Current	A
	51324	PV5 Current	A
	51326	PV6 Current	A
	51333	PV1 Power	W
	51334	PV2 Power	W
	51335	PV3 Power	W

	51336	PV4 Power	W
	51337	PV5 Power	W
	51338	PV6 Power	W
	51347	PV1 Yield Today	Wh
	51348	PV2 Yield Today	Wh
	51349	PV3 Yield Today	Wh
	51350	PV4 Yield Today	Wh
	51351	PV5 Yield Today	Wh
	51352	PV6 Yield Today	Wh
	51339	PV1 Total Yield	Wh
	51340	PV2 Total Yield	Wh
	51341	PV3 Total Yield	Wh
	51342	PV4 Total Yield	Wh
	51343	PV5 Total Yield	Wh
	51344	PV6 Total Yield	Wh

Appendix 1: Device Type Dictionary Definitions

device_type	Device Type Name (CN)	Device Type Name (EN)
1	逆变器	Inverter
2	集装箱	Container
3	并网点	Grid-Connection Point
4	汇流箱	Combiner Box
5	环境监测仪	Meteo Station
6	变压器	Transformer
7	电表	Meter
8	UPS	UPS
9	数据采集器	Data Logger
10	组串	String
11	电站	Plant
12	线路保护	Circuit Protection
13	解列装置	Splitting Device
14	储能逆变器	Energy Storage System
15	采集设备	Sampling Device
16	EMU	EMU
17	单元	Unit
18	温湿度传感器	Temperature and Humidity Sensor
19	智能配电柜	Intelligent Power Distribution Cabinet
20	显示设备	Display Device
21	交流配电柜	AC Power Distributed Cabinet
22	通信模块	Communication Module
23	系统BMS	System-BMS
24	阵列BMS	Array-BMS
25	直流-直流	DC-DC

26	能量管理系统	Energy Management System
27	跟踪系统	Tracking System
28	风能变流器	Wind Energy Converter
29	SVG	SVG
30	PT柜	PT Cabinet
31	母线保护	Bus Protection
32	清扫机器人	Cleaning Device
33	直流屏	Direct Current Cabinet
34	公用测控	Public Measurement and Control
37	储能变流器	Energy Storage System
44	电池簇管理单元	Battery Cluster Management Unit
45	本地控制器	Local Controller
52	电池系统控制器	Battery System Controller

Appendix 2: API Error Code Definitions

Error Code Name	Type	Error Code
success	Call successful	1
error	Internal service exception	-1
er_unknown_exception	Unknown exception	000
er_missing_parameter:appkey	appkey cannot be empty	001
er_missing_parameter:token	token cannot be empty	002
er_missing_parameter:sys_code	sys_code cannot be empty	003
er_invalid_appkey	Invalid appkey	E00000
er_api_service_has_expired	API service expired	E00001
er_parameter_decrypt_error	Parameter decryption exception	E00002
er_token_login_invalid	The token is invalid or has expired	E00003
er_month_call_api_times_upper_limit	Max API calls reached	E998
er_hour_call_api_times_upper_limit	Max API calls reached	E999
er_missing_parameter	Required parameter missing	009
er_parameter_value_invalid	Parameter value invalid	010
er_sql_exception	SQL exception	011
Unauthorized access	Access has not been authorized	E900
Call too frequently	The call frequency is too high	E901
Abnormal network environment	IP address change frequency is too high	E903
Request is not encrypted	The request has not been encrypted	E902
Missing parameter in request header: x-random-secret-key	The x-random-secret-key is missing from the request header	E904
AES decryption exception	There was an issue with AES decryption	E905
RSA decryption exception	There was an issue with RSA decryption	E906
AES random secret key length must be 16	The length of the AES random secret key must be 16	E907
Missing key parameter: api_key_param	The api_key_param is missing	E908
Invalid parameter format: nonce [32-bit string of numbers and letters]	The parameter format is invalid: nonce [32-bit string of numbers and letters]	E909
Repeated request	The request nonce must be regenerated for repeated requests	E910

Missing parameter in request header: x-access-key	The x-access-key is missing from the request header	E911
Illegal x-access-key	The x-access-key is invalid	E912
Expired request	The request has expired, the time difference between the timestamp (UTC+0 UNIX) in the request and the server time is too high	E913
Mismatched appkey and x-access-key	The appkey and access-key do not match	E914
Login too frequently	Too many repeated login attempts	E916

```
import org.apache.http.client.methods.CloseableHttpResponse;
import org.apache.http.client.methods.HttpPost;
import org.apache.http.entity.StringEntity;
import org.apache.http.impl.client.CloseableHttpClient;
import org.apache.http.impl.client.HttpClients;
public class UserTest{
    Public static void main(String[] args)
    {
        CloseableHttpClienthttpclient = HttpClients.createDefault();
        String url = "https://API domain name which is provided by iSolarCloud/xxx/xxx";
        HttpPosthttppost = new HttpPost(url);
        try
        {
            // Header
            httppost.addHeader("sys_code", "901");
            HashMap<String, Object>req = new HashMap<String, Object>();
            // Public parameter
            req.put("appkey", "authorized appkey");
req.put("token", "token returned by login API");
            // Service
            req.put("service", "service name");
            req.put("parameter 1", "parameter value");
            String jsonStr = com.alibaba.fastjson.JSON.toJSONString(req);
            System.out.println("send json-<" + jsonStr.toString());
            StringEntitystrEntity = new StringEntity(jsonStr);
            strEntity.setContentType("application/json");
            httppost.setEntity(strEntity);
            CloseableHttpResponse response = httpclient.execute(httppost);
            try
            {
                HttpEntity entity = response.getEntity();
                InputStreaminputStream = entity.getContent();
                InputStreamReaderinputStreamReader = new InputStreamReader(
                        inputStream, "UTF-8");
                BufferedReader reader = new BufferedReader(inputStreamReader);
                StringBuilder result = new StringBuilder();
                String s;
                while (((s = reader.readLine()) != null))
                {
                    result.append(s);
                }
                reader.close();
                System.out.println("receive json-<" + result.toString());
```

```
            }
            finally
            {
                response.close();
            }
        }
        catch (Exception e)
        {
            e.printStackTrace();
        }
        finally
        {
            try
            {
                httpclient.close();
            }
            catch (IOException e)
            {
                e.printStackTrace();
            }
        }
    }
}
```

```
1. Header encapsulation
String publicKey = "xxxxx" ;// publicKey assigned by iSolarCloud
// publicEncrypt methods in Appendix 3
String x-random-secret-key = publicEncrypt("A123456zA123456z", publicKey) ;//x-random-secret-key could be different i
String x-access-key = "i71w7tskmns5********i3b8zqncvay3" ;//accessKey assigned by iSolarCloud

2. Body encapsulation
// Use login API as an example，Besides request body parameter  in 2.1, api_key_param is also in need：
//nonce is a 32 bit random string of letters and numbers, it should be different in every call;
//timestamp is UNIX time stamp in milliseconds，if error code E913 is returned,
// Use GET method to call  https://API domain name which is provided by iSolarCloud/timestamp
String requestBody =
{
    "api_key_param": {
        "nonce": "cb360459bd624c6ab15308c4b6847856",
        "timestamp": "1616725497384"
    },
    "appkey": "***********",
    "login_type": "1",
    "user_account": "******",
    "user_password": "******"
};

3. Request call
CloseableHttpClient httpclient = HttpClients.createDefault();
String url = "https://API domain name which is provided by iSolarCloud/v1/userService/login";
HttpPost httppost = new HttpPost(url);
// Request header
```

```
httppost.addHeader("x-random-secret-key", x-random-secret-key);
httppost.addHeader("x-access-key", x-access-key);
...... //Set other request header
// Encryption methods in Appendix 4
String encryptedRequestBody = encrypt(requestBody, "A123456zA123456z") ;
StringEntity strEntity = new StringEntity(encryptedRequestBody);
strEntity.setContentType("application/json");
httppost.setEntity(strEntity);
CloseableHttpResponse response = httpclient.execute(httppost);
HttpEntity entity = response.getEntity();
InputStream inputStream = entity.getContent();
InputStreamReader inputStreamReader = new InputStreamReader(inputStream, "UTF-8");
BufferedReader reader = new BufferedReader(inputStreamReader);
StringBuilder responseBody = new StringBuilder();
String s = null;
while (((s = reader.readLine()) != null))
{
   responseBody.append(s);
}
reader.close();
// Decryption methods in Appendix 5
String decryptedResponseBody = decrypt(responseBody, "A123456zA123456z") ;
```